Learn Citaville Institute - Brochure

Python Software Engineering

About the Course Program

This course builds strong foundations in Python for software development. Learners will progress from basic syntax to advanced object-oriented programming, data structures, and backend development. It emphasizes practical projects for real-world applications.

Delivery: Virtual instructor-led sessions with coding assignments

Certification Requirement: Completion of projects and minimum attendance. Earn a minimum passing grade of 70% in each Course Topic assessment

Career Paths: Python Developer, Backend Engineer, Data Engineer, Software Engineer

Recommended Study Time: Minimum of an hour personal daily study time during week days and 5 - 6 hours intensive training during weekend

Program Duration: 8 - 12 weeks

Who Can Attend?

This program is designed for:

- Beginners and intermediate learners seeking coding skills

- Professionals aiming for software engineering careers
- Students transitioning to tech roles

Pre-requisites

- Basic computer literacy
- Prior programming knowledge is helpful but not required

Detailed Course Overview & Topics

- **Module 1: Introduction to Software Engineering:** Explore fundamental concepts of the software development life cycle (SDLC), agile methodology, and software architecture.
- **Module 2: Python Fundamentals:** Hands-on practice with Python syntax, data structures, and object-oriented programming (OOP) principles.
- **Module 3: Version Control with Git & GitHub:** Learn to use Git and GitHub for collaborative development and version management.
- **Module 4: Practical Project - Building a Command-Line Tool:** Apply your Python and Git skills to build a simple, functional command-line tool.
- **Module 5: Web Development with Flask/Django:** Dive into web frameworks, building a RESTful API and a basic web application.
- **Module 6: Testing and Quality Assurance:** Learn to write and run unit and integration tests to ensure software quality.
- **Module 7: Project-Based Learning:** Work on a capstone project to apply all the learned concepts, such as a web-based task management system.

On Completion of this Course, You'll Walk Away With:

1

Proficiency in Python programming

2

Portfolio-ready coding projects

3

Foundation for advanced software engineering

4

Ability to develop, test and deploy Python applications with best practices.